HG-Bench: A Benchmark for Multi-Page Handwritten Answer-Region Grounding in Automated Homework Assessment

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Abstract

Automated homework assessment requires not only recognizing student answers but accurately locating complete answers and intermediate reasoning steps in noisy, multi-page handwritten work. We address the missing evaluation setting of *page-aware, two-level answer-region grounding*: given homework page images, a model must localize complete answer regions and their ordered step-level subregions. We introduce **HG-Bench**, comprising 500 human-annotated K–12 homework samples curated from a 1,489,278-image source pool, with hierarchically linked question- and step-level boxes. Its page-aware protocol separately measures complete-answer localization ($\mathcal{F}_A$) and step-level decomposition ($\mathcal{F}_S^{\mu}$), distinguishing structured spatial grounding from visible-text parsing. Across frontier closed-source and open-weight VLMs, no zero-shot system exceeds 55.22% on $\mathcal{F}_A$ or 48.22% on $\mathcal{F}_S^{\mu}$, whereas a GLM-4.6V 9B reference fine-tuned on ∼10k in-domain examples reaches 74.97/72.26. These results expose step-level handwritten grounding as a concrete capability gap and provide a reproducible benchmark, evaluation protocol, and trained reference for future work on automated homework assessment.

1 Introduction

Automated homework assessment is increasingly deployed in educational settings to relieve teacher workload and to improve grading consistency. The first stage of every such pipeline is *spatial localization*: the system must identify, on each scanned page, where a student has written each answer and, for multi-step problems, where each constituent step of the derivation lies. Downstream optical character recognition, grading, and feedback modules consume these regions, so the accuracy of the final grade is bounded above by the accuracy of upstream grounding.

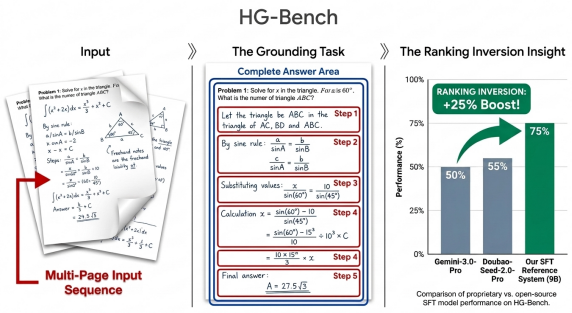


Figure 1: **HG-Bench at a glance.** HG-Bench takes a sequence of multi-page handwritten homework images as input and requires models to produce page-aware, two-level grounding outputs: complete answer regions for each question and ordered step-level boxes for multi-step solutions. This setting tests whether VLMs can localize the spatial structure of student reasoning, not only recognize final answers.

Research gap. Despite rapid progress on referring-expression grounding in natural images (Kazemzadeh et al., 2014; Plummer et al., 2015) and on document understanding (Mathew et al., 2021; Masry et al., 2022; Mathew et al., 2022), no public benchmark measures grounding on real handwritten student work. Natural-image benchmarks evaluate a single referent per query in editorial photographs; document-AI benchmarks recover text content but do not require spatially correct, hierarchically structured region outputs. Educational AI benchmarks such as MathVista (Lu et al., 2024) and MathVerse (Zhang et al., 2024) evaluate problem solving assuming the relevant content has already been identified. The capability most relevant to real assessment pipelines—ordered, two-level, page-aware grounding on noisy multi-page handwritten scans—remains unmeasured.

Core challenges. Real homework scans are markedly harder than the inputs of any prior grounding benchmark along four axes simultaneously: (i) **multi-page** samples with page-aware co-

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ordinate semantics; (ii) **handwriting** with irregular line spacing, hand shadow, perspective skew, and student-specific stroke styles; (iii) **two-level structure**, requiring both per-question answer regions and ordered per-step sub-regions under hierarchical containment; and (iv) **long-tail layout heterogeneity** across subjects (e.g., fraction derivations and matrices in mathematics versus dense paragraphs in language subjects). A benchmark that fails to expose any one of these axes will overstate the readiness of current vision–language models (VLMs) for assessment deployment.

Our approach. We address this gap with two coordinated artefacts. First, HG-Bench: a curated, stratified, human-annotated test set of 500 multi-page samples covering all four challenge axes, together with a page-aware evaluation protocol that decouples question-level localization ($\mathcal{F}_A$) from step-level decomposition ($\mathcal{F}_S^\mu$). Second, a lightweight *reference fine-tuned system* obtained by single-stage supervised fine-tuning of an open-weight 9B VLM on ∼10k in-domain examples, included as a trained reference point rather than as a SOTA submission—it verifies that the benchmark is learnable and quantifies a lower bound on dedicated-pipeline performance.

Key empirical findings. We observe a pronounced ranking inversion across paradigms (Tab. 2): the best closed-source results are 55.22% on $\mathcal{F}_A$ (Doubao-Seed-2.0-Pro, evaluated 2026-04-01) and 48.22% on $\mathcal{F}_S^\mu$ (Gemini-3-Pro-Preview). The reference SFT system reaches $74.97/72.26$. The headline gap is largest on $\mathcal{F}_S^\mu$ ($+24.04$ absolute), confirming that step-level structured grounding—rather than coarse answer-region detection—is the central capability HG-Bench measures. Crucially, scale does not close the gap: the 397B-parameter Qwen3.5-397B-A17B (Qwen Team, 2026) attains only $42.71/18.15$, lower than several smaller closed APIs.

Contributions.

- **HG-Bench**, the first benchmark for two-level, page-aware answer-region grounding on multi-page handwritten K–12 homework, comprising 500 human-annotated samples curated from a 1.49M-image source pool.
- A **page-aware evaluation protocol** that decouples the question-level macro metric $\mathcal{F}_A$ from the step-level micro metric $\mathcal{F}_S^\mu$ computed over step-bearing pages, correcting for the structural imbalance of multi-step problems across samples.
- A **systematic evaluation** of nine frontier vision–language systems—closed-source APIs (GPT-5.4, Claude-Sonnet-4.6, Doubao-Seed-2.0-Pro, Gemini-3-Pro-Preview) and open-weight models (Qwen3.5-397B-A17B, GLM-5V-Turbo, Kimi-K2.5, GLM-4.6V 9B) (OpenAI, 2026; Anthropic, 2026; ByteDance Seed Team, 2026; Google, 2025; Qwen Team, 2026; GLM-V Team, 2026; Kimi Team, 2026; GLM-V Team, 2025)—establishing that step-level grounding is the dominant capability gap and that parameter count alone does not close it.
- A **reference fine-tuned system** obtained by single-stage SFT of GLM-4.6V 9B on ∼10k in-domain examples, which surpasses every evaluated closed-source baseline without any reinforcement-learning stage. We release the checkpoint as a lower-bound reference for future HG-Bench submissions.

2 Related Work

Visual grounding. Referring-expression grounding is benchmarked on natural-image datasets including RefCOCO (Kazemzadeh et al., 2014; Yu et al., 2016; Mao et al., 2016), Flickr30K Entities (Plummer et al., 2015), and Visual Genome (Krishna et al., 2017). These benchmarks mainly pair each query with a single natural-image referent. Detection-oriented systems such as GLIP (Li et al., 2022a) and Grounding-DINO (Liu et al., 2024a) improve localization, while Shikra (Chen et al., 2023) and Ferret (You et al., 2024) integrate referring into multimodal dialogue. Related tasks broaden the query and output spaces: TOIST maps action affordances to instance masks (Li et al., 2022b), Touch-Line Transformer combines language with gestural geometry (Li et al., 2023a), and GUI work links element localization to downstream actions (Li et al., 2026). Dense prediction spans language-guided formulations (Yao et al., 2026a) and saliency-aware open-vocabulary segmentation (Yao et al., 2026b). Extensions also couple localization with cross-video procedures (Ye et al., 2026), evidence-anchored attention (Hu et al., 2026), preference-optimized spatial reasoning (Shen et al., 2025), and compositional 3D un-

derstanding (Yu et al., 2025a). Mechanistic analyses study VLM grounding and hallucination circuits (Liu et al., 2026a) as well as shared decode-time decision subspaces (Shao et al., 2026), while structure-aware segmentation (Zhang et al., 2026a) and human-aligned layout evaluation (Lu et al., 2025) emphasize spatial organization. HG-Bench differs by requiring page-aware, hierarchical question and ordered step regions on handwritten scans.

Grounding capabilities of recent VLMs. Bounding-box generation is supported by closed-source VLMs such as GPT-4V/4o (OpenAI, 2023), Gemini (Gemini Team et al., 2023), Claude (Anthropic, 2024), Doubao (ByteDance Seed Team, 2025), and Kimi (Kimi Team, 2026), and by open-weight families including Qwen-VL/Qwen2.5-VL (Wang et al., 2024a; Bai et al., 2025), InternVL/InternVL2.5 (Chen et al., 2024a,b), CogVLM2 (Wang et al., 2024b), MiniCPM-V (Yao et al., 2024), Florence-2 (Xiao et al., 2024), LLaVA-NeXT (Liu et al., 2024b), DeepSeek-VL2 (Wu et al., 2024), Phi-3-Vision (Abdin et al., 2024), and GLM-V (GLM-V Team, 2025). These systems inherit visual–linguistic priors from backbones such as CLIP (Radford et al., 2021) and ALIGN (Jia et al., 2021); Mixture-of-Experts sparsity further scales open-weight capacity (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024; Dai et al., 2024), but does not itself ensure fine-grained grounding. Adjacent vision–language–action work studies informative-frame selection and generalist–specialist adaptation (Yu et al., 2026a,b) as well as progress-aware affordance reasoning for long-horizon interaction (Liu et al., 2026b). Cross-modal asynchronous fusion instead separates modality-specific representation learning from semantic interaction (Zhang et al., 2026b). Despite these advances, none has been systematically evaluated on page-aware, hierarchical grounding in handwritten K–12 work, and none solves HG-Bench.

Educational AI benchmarks and benchmark methodology. Mathematical and scientific benchmarks—MathVista (Lu et al., 2024), MathVerse (Zhang et al., 2024), We-Math (Qiao et al., 2024), MATH-Vision (Wang et al., 2024c), MathSpatial (Lu et al., 2026), and OlympiadBench (He et al., 2024)—measure reasoning after the relevant visual content has been identified. Broader suites such as MMBench (Liu et al., 2024c), SEED-Bench (Li et al., 2023b), MMMU (Yue et al., 2024), Video-MME-Logical (Kwan et al., 2026), HELM (Liang et al., 2023), and BIG-Bench (Srivastava et al., 2023) target general capability coverage, while cited-RAG diagnostics test whether retrieved evidence warrants a claim (Qian et al., 2026). Neither category measures region-level grounding on handwritten answers; inter-annotator agreement using Cohen's κ (Cohen, 1960) or Fleiss' κ (Fleiss, 1971) also remains under-reported in this domain.

3 Task Formulation

Problem definition. The goal of the task is to evaluate a model's ability to accurately localize student-written answers at both the question and the step level, which is essential for automated grading and trace-of-reasoning analysis in multi-page handwritten homework.

Inputs and outputs. The input to the system is an ordered sequence of homework page images $\{\mathbf{I}_p\}_{p=1}^{P}$ accompanied by metadata specifying each page's pixel dimensions. The expected output is a JSON array $\{q_i\}_{i=1}^{N}$, where each element q_i corresponds to one question and contains:

- a fixed type field `complete_answer_box`;
- a page index $p_i \in \{1, \ldots, P\}$;
- a question-level bounding box $\mathbf{b}_i \in [0, 1000]^4$ in xyxy format, enclosing the full handwritten answer to the question;
- an optional ordered list of step boxes $\{s_{i,j}\}_{j=1}^{K_i}$, each carrying a `step_id` (one-indexed in the student's writing order) and its own box $\mathbf{b}_{i,j}$.

All question and step boxes must be emitted in the order of the student's answers.

Two-level box semantics. Question-level boxes (occasionally referred to as "title boxes" in the annotation tool) localize the complete handwritten answer region of each question and support per-item scoring and partial-credit attribution. Step-level boxes further decompose multi-step solutions and multi-blank responses into ordered sub-regions, enabling step-level grading and trace-of-reasoning analysis. Each $\mathbf{b}_{i,j}$ must be fully contained within its parent $\mathbf{b}_i$, enforcing hierarchical consistency. When a region is partially missing or ambiguous,

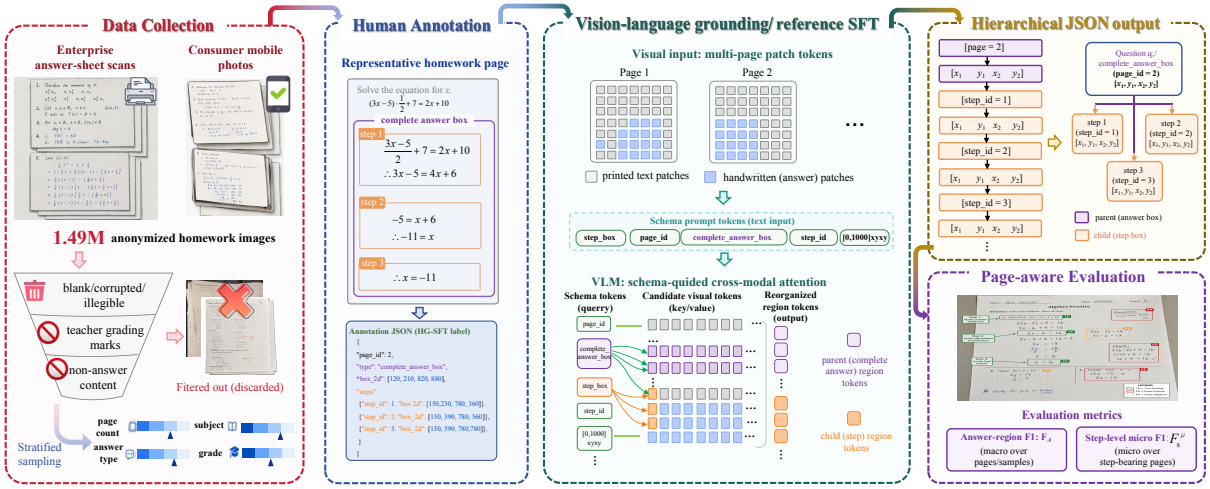


Figure 2: **HG-Bench data engine.** The pipeline proceeds in four stages. **(I) Collection.** 1,489,278 anonymized homework images are gathered from two complementary channels—enterprise B2B answer-sheet scans and consumer photographs from a public-facing consumer homework-photo application—to expose both controlled-capture and in-the-wild distributions. **(II) Filtering.** Images are filtered for usability (corruption, blank, illegibility), for absence of teacher grading marks, and for student-answer content; multi-stage automatic checks are followed by manual verification. **(III) Stratified annotation.** 10,420 samples are annotated under a two-level boxing protocol (question-level + ordered step-level) with strict hierarchical containment; annotators use a custom tool with keyboard-shortcut box drawing. **(IV) Verification and stratified split.** Every sample is reviewed at least once; ambiguous items are escalated to a lead annotator. The annotated pool is stratified along subject, grade, page count, and answer type, yielding a 500-sample test split (HG-Bench) and a 9,920-sample training pool (HG-SFT) used by the reference system in Sec. 6.2.

models should predict the best-fit bounding box while preserving this containment rule.

Coordinate convention. Coordinates are normalized to the $[0, 1000]$ xyxy format by default and can be denormalized to pixel values using per-page metadata. The protocol also supports yxyx and pixel-coordinate variants via configuration. Models that emit polygons are evaluated via the minimum enclosing axis-aligned rectangle. All coordinates must tightly enclose student-written content and exclude printed text and teacher annotations.

4 HG-Bench

4.1 Source Pool

HG-Bench is derived from a large pool of 1,489,278 anonymized student homework images, comprising real online homework scans and ink-screen captures spanning multiple subjects and grade levels. From this raw pool, we curate a high-quality annotated set of 10,420 valid samples from two representative collection channels:

- **Enterprise channel** (6,765 samples), drawn from formal answer sheets and standardized exam grading, characterized by multi-page samples and a high proportion of multi-step problems;
- **Consumer channel** (3,655 samples), drawn from homework photos uploaded by general users of a public-facing consumer homework-photo application, typically single-page images with a more balanced distribution of question types and in-the-wild capture variation.

The annotated pool is then partitioned into a 500-sample held-out test set (**HG-Bench**) and a 9,920-sample training pool (**HG-SFT**), with 250 samples held out from each channel. The test and training pools are disjoint by construction; we additionally verify disjointness with perceptual hashing (Sec. 6.2). All personally identifying information was removed before inclusion in either pool.

4.2 Sampling Strategy

To ensure that the benchmark is representative of real-world homework, we perform stratified sampling along four axes: subject, grade level, page count per sample, and answer type. This procedure yields the 500 benchmark samples and is designed to capture the long tail of layouts and problem

complexities encountered in practice. Detailed per-stratum counts appear in Table 1 and in Figure 3.

4.3 Annotation Protocol

The annotation protocol specifies which pages to skip and prescribes the procedure for drawing the two levels of bounding boxes.

Skip rules. A page is skipped if (i) the image is unusable (corrupted, blank, or illegible); (ii) teacher grading marks (✓, ×, written scores, etc.) are present; or (iii) all student answers are non-conventional (drawings, doodles, off-task content). Pages containing red-pen marks not associated with grading are retained and annotated normally.

Box drawing. Each region of student handwriting is enclosed by an axis-aligned bounding box. Single-answer questions receive one question-level box. Multi-step solutions and multi-blank responses additionally receive an ordered set of step-level boxes, each tightly enclosing one step's handwriting and never splitting a single handwritten line across multiple boxes. Informal scratch work is not boxed.

Tagging. Each box carries the local question number and the parent-number hierarchy, separated by "/". Question type is tagged from a fixed inventory: choice, fill, judgment, solve (including computation), drawing, short-answer, writing. Step IDs are integers assigned in the order of the student's writing. All punctuation in tags follows Chinese typographic conventions.

4.4 Annotation Workflow

Annotations followed a two-stage protocol. **12** trained annotators drew question- and step-level boxes with a custom shortcut-based tool, and **5** senior reviewers independently accepted each annotation or returned it for revision; ambiguous cases were escalated to a lead annotator. We measured inter-annotator agreement (IAA) on 50 randomly sampled test samples annotated by two independent annotators before QC revisions. For localization, the annotations achieved a mean IoU of 0.86, with 85% of boxes matched at IoU $= 0.5$, and Cohen's $\kappa = 0.83$ for binary box-matching consistency. For discrete annotations—question type, step containment, and step ordering—Fleiss' κ was 0.81. These results indicate substantial agreement (Cohen, 1960; Fleiss, 1971) and support the reliability of HG-Bench.

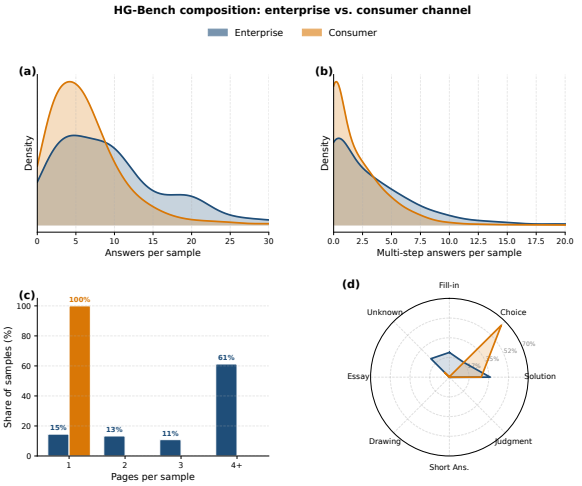


Figure 3: **HG-Bench composition.** (a) Answers per sample follow a long-tail distribution in both enterprise and consumer channels. (b) Multi-step problems are over-represented in the enterprise channel, consistent with formal exam scenarios. (c) Page count per sample: enterprise samples are predominantly multi-page; consumer samples are predominantly single-page. (d) Question-type composition: fill-in-the-blank dominates the enterprise channel, while the consumer channel is more uniformly distributed across choice, fill, solve, and short-answer types. The 500-sample benchmark is stratified along these four axes to remain representative of both channels.

Table 1: HG-Bench summary statistics.

Statistic	Value
Samples	500
Subjects	5
Mean pages per sample	1.8
Mean question boxes per page	5.2
Mean step boxes per question (when present)	2.9
Fraction of step-bearing pages	0.34

4.5 Dataset Statistics

We summarize the annotated pool and HG-Bench test set in Figure 3 and Table 1. The data reflect realistic homework scenarios: both channels show long-tailed answer counts, while enterprise samples contain more multi-step problems, more multi-page submissions, and more fill-in-the-blank items. Consumer samples are mostly single-page and more balanced across choice, fill-in-the-blank, solve, and short-answer questions. The 500 HG-Bench samples are stratified to preserve these patterns, making the benchmark representative and challenging for both question-level and step-level grounding.

5 Evaluation Protocol

5.1 Page-Aware Box Matching

Unlike standard grounding tasks that flatten all bounding boxes across an entire document, our evaluation operates strictly at the page level to preserve the multi-page layout structure.

Within each page, we perform greedy one-to-one matching between ground-truth boxes $\mathcal{G}$ and predicted boxes $\mathcal{P}$ under an Intersection-over-Union (IoU) constraint. Specifically, we filter out all candidate pairs $(g, p) \in \mathcal{G} \times \mathcal{P}$ with $\mathrm{IoU}(g, p) < 0.5$. The remaining pairs are sorted in descending order of IoU and assigned greedily, so that each box participates in at most one match. This localized matching yields per-page true-positive (TP), false-positive (FP), and false-negative (FN) counts.

5.2 Reported Metrics

Following page-level matching, performance is aggregated into two primary dataset-level localization metrics; we additionally report four supplementary metrics defined in Appendix D to characterize step-decomposition robustness ($\mathcal{F}_S^M$) and parse-time reliability ($\mathrm{Succ}\%$, $\bar{\mathcal{S}}$, $\mathrm{Rep}\%$).

- **Answer-region F_1 ($\mathcal{F}_A$).** This metric evaluates the localization quality of question-level answer regions. We compute the standard F_1 score per page from its local TP, FP, and FN counts, average within each multi-page sample, and report the macro-average across all successfully evaluated samples:

$$\mathcal{F}_A = \frac{1}{|\mathcal{S}_{\text{succ}}|} \sum_{s \in \mathcal{S}_{\text{succ}}} \left(\frac{1}{|\mathcal{M}_s|} \sum_{m \in \mathcal{M}_s} \mathrm{F}_1(s, m) \right) \times 100 \quad (1)$$

  where $\mathcal{S}_{\text{succ}}$ is the set of successfully evaluated samples, $\mathcal{M}_s$ is the set of pages within sample s, and $\mathrm{F}_1(s, m)$ is the box-level F_1 on the m-th page of sample s.

- **Step-level micro F_1 ($\mathcal{F}_S^\mu$).** Unlike question-level answer regions, which are present on every page, step-level boxes exhibit a highly sparse and imbalanced distribution: many pages contain no steps at all. To eliminate the evaluation bias induced by this imbalance, $\mathcal{F}_S^\mu$ is computed by micro-aggregation restricted to step-bearing pages. We accumulate step-level TP, FP, and FN counts globally across all pages whose ground truth contains at least one step box and compute a single unified F_1 score.

6 Benchmarking and Experimental Analysis

All baseline VLMs use the same prompt template (Appendix C), which fixes the JSON schema, normalized $[0, 1000]$ coordinates, and page-aware sequence-preserving output format. We parse outputs with a format-tolerant JSON parser and issue one format-reminder retry after structural failures; persistent failures are marked with `FAIL_STR` (`success` = *False*). Because models differ in coordinate conventions, we auto-detect box-axis ordering per model on a small held-out calibration slice before evaluation.

6.1 Evaluated Baselines

We evaluate two cohorts of frontier vision–language systems:

- **Closed-source frontier APIs.** GPT-5.4 (OpenAI, 2026), Claude-Sonnet-4.6 (Anthropic, 2026), Doubao-Seed-2.0-Pro (ByteDance Seed Team, 2026) (the official `260215` release and a later evaluation on 2026-04-01), and Gemini-3-Pro-Preview (Google, 2025), called through their official endpoints under default decoding.

- **Open-weight baselines.** Qwen3.5-397B-A17B (Qwen Team, 2026) (a Mixture-of-Experts model activating 17B parameters per token), GLM-5V-Turbo (GLM-V Team, 2026), Kimi K2.5 (Kimi Team, 2026), and the GLM-4.6V 9B base (GLM-V Team, 2025).

We additionally report **GLM-4.6V-9B + HG-SFT**, a reference fine-tuned system trained on the HG-SFT pool. This model is included as a trained reference point rather than a zero-shot baseline, allowing us to test whether HG-Bench is learnable with a modest amount of in-domain supervision.

6.2 Reference SFT System

To verify learnability and provide a reproducible lower-bound reference, we fine-tune GLM-4.6V 9B on the 9,920-sample HG-SFT training pool using single-stage supervised fine-tuning. Structured agent distillation separates reasoning and action segments to preserve structured agent behavior in smaller models (Liu et al., 2026c). In a different structured-prediction domain, reverse-distilled fine-tuning internalizes decomposed reasoning trajectories for interpretable Text-to-SQL (Guan et al.,

Table 2: Main results on HG-Bench ($N = 500$ samples for every model). $\mathcal{F}_A$: macro answer-region F_1 averaged across samples and pages. $\mathcal{F}_S^{\mu}$: micro step-level F_1 aggregated over step-bearing pages. $\mathcal{F}_S^{M}$: macro step-level F_1 averaged over step-bearing samples (Appendix D). Succ%: parse success rate. $\bar{\mathcal{S}}$: unified composite score averaged over all 500 samples (failed parses count as 0). Rep%: fraction of outputs containing repeated content (lower is better). Best results are shown in bold, second-best underlined.

Model	$\mathcal{F}_A$	$\mathcal{F}_S^{\mu}$	$\mathcal{F}_S^{M}$	Succ%	$\bar{\mathcal{S}}$	Rep%
Closed-source frontier APIs						
GPT-5.4	14.91	1.55	1.38	100.0	8.12	0.4
Claude-Sonnet-4.6	16.83	1.63	1.21	99.2	8.76	2.8
Doubao-Seed-2.0-Pro (260215)	52.65	44.78	<u>42.59</u>	49.4	21.22	**0.0**
Doubao-Seed-2.0-Pro (evaluated 2026-04-01)	<u>55.22</u>	40.11	34.88	99.8	<u>42.70</u>	<u>0.2</u>
Gemini-3-Pro-Preview	50.90	<u>48.22</u>	37.58	100.0	42.33	6.0
Open-weight baselines						
Qwen3.5-397B-A17B	42.71	18.15	17.73	94.2	32.73	4.0
GLM-5V-Turbo	46.69	26.29	23.78	100.0	40.10	0.4
Kimi K2.5	31.21	7.42	7.21	79.4	20.18	1.0
GLM-4.6V 9B (base)	34.15	7.65	4.60	100.0	29.46	3.8
Reference SFT system						
GLM-4.6V-9B + HG-SFT	**74.97**	**72.26**	**48.25**	100.0	**71.53**	0.8
Δ over best prior	↑ 19.75	↑ 24.04	↑ 5.66	–	↑ 28.83	–

2026); our reference system instead directly supervises hierarchical question- and step-level regions without a teacher–student distillation stage. The training uses no reinforcement learning, no synthetic continued pre-training, and no out-of-domain data mixing, so improvements over zero-shot baselines can be attributed to targeted in-domain supervision rather than to a stronger foundation model or additional training stages. Full details on the base checkpoint, train–test deduplication, data composition, and optimization recipe are provided in Appendix B.

6.3 Main Results

Table 2 reports results across all baselines and the reference SFT system.

Frontier VLMs plateau well below a trained reference, and scale alone does not close the gap. No zero-shot baseline exceeds 55.22 on $\mathcal{F}_A$ or 48.22 on $\mathcal{F}_S^{\mu}$, whereas the ∼10k-example reference reaches 74.97/72.26, gains of 19.75/24.04 points. Despite 397B parameters, Qwen3.5-397B-A17B scores only 42.71/18.15, trailing smaller closed APIs on $\mathcal{F}_A$ and GLM-5V-Turbo on $\mathcal{F}_S^{\mu}$. Thus, scale alone does not solve fine-grained ordered localization, and HG-Bench remains far from saturated.

Step-level grounding is the dominant capability gap. Every zero-shot model drops markedly from $\mathcal{F}_A$ to $\mathcal{F}_S^{\mu}$. Kimi K2.5 and GLM-4.6V 9B have single-digit step scores, GPT-5.4 and Claude-Sonnet-4.6 score only 1.55 and 1.63, and the strongest closed API falls from 55.22 to 40.11. The reference narrows this gap to 2.71 points (74.97 vs. 72.26), supporting targeted supervision and motivating $\mathcal{F}_S^{\mu}$ as the primary ranking metric.

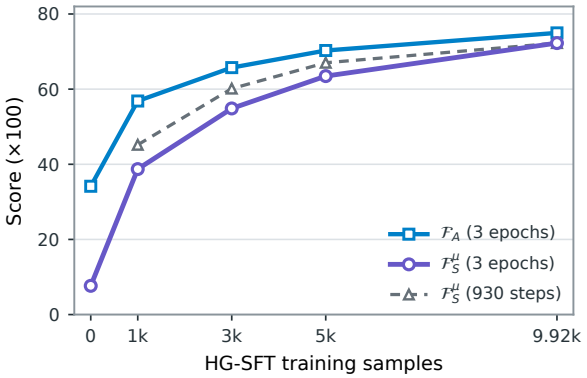


Figure 4: **HG-SFT data scaling is monotonic and not saturated.** Solid lines use three training epochs; the dashed line fixes all runs to 930 optimization steps.

Performance improves with unique in-domain supervision. Figure 4 summarizes nested subsets stratified by channel, subject, page count, and answer type. With three epochs, scaling from 1k to 9.92k samples raises $\mathcal{F}_A$ from 56.84 to 74.97 and $\mathcal{F}_S^{\mu}$ from 38.72 to 72.26; even the 5k-to-9.92k increment adds 8.79 step-level points. With training fixed at 930 steps, $\mathcal{F}_S^{\mu}$ still rises monotonically (45.18, 60.14, 67.02, 72.26), confirming that additional unique examples—not extra optimization—drive the gain.

6.4 Subject and Page-Count Breakdown

Subject breakdown. Across the evaluated subjects, zero-shot baselines score lowest on Mathematics and Science. We attribute this to the nonlinear layout of STEM solutions—fraction derivations, matrices, spatial scratch notes—which violates the predominantly left-to-right, top-to-bottom prior of generic VLM training. Language subjects (Chinese, English) follow a more regular linear sequence, making question-level bounding ($\mathcal{F}_A$) easier; even so, fine-grained step tracking ($\mathcal{F}_S^{\mu}$) degrades on long-form answers with dense interlocking paragraphs, indicating that the bottleneck is layout density rather than subject identity per se.

Page-count scaling. We slice the test set by input page count (1, 2, 3+). Zero-shot baselines retain coordinate formatting on single-page inputs but exhibit monotonic decay in step-level metrics as page count grows; across baselines, $\mathcal{F}_S^{\mu}$ drops by over 25% on average from 1 to 3+ pages. The failure

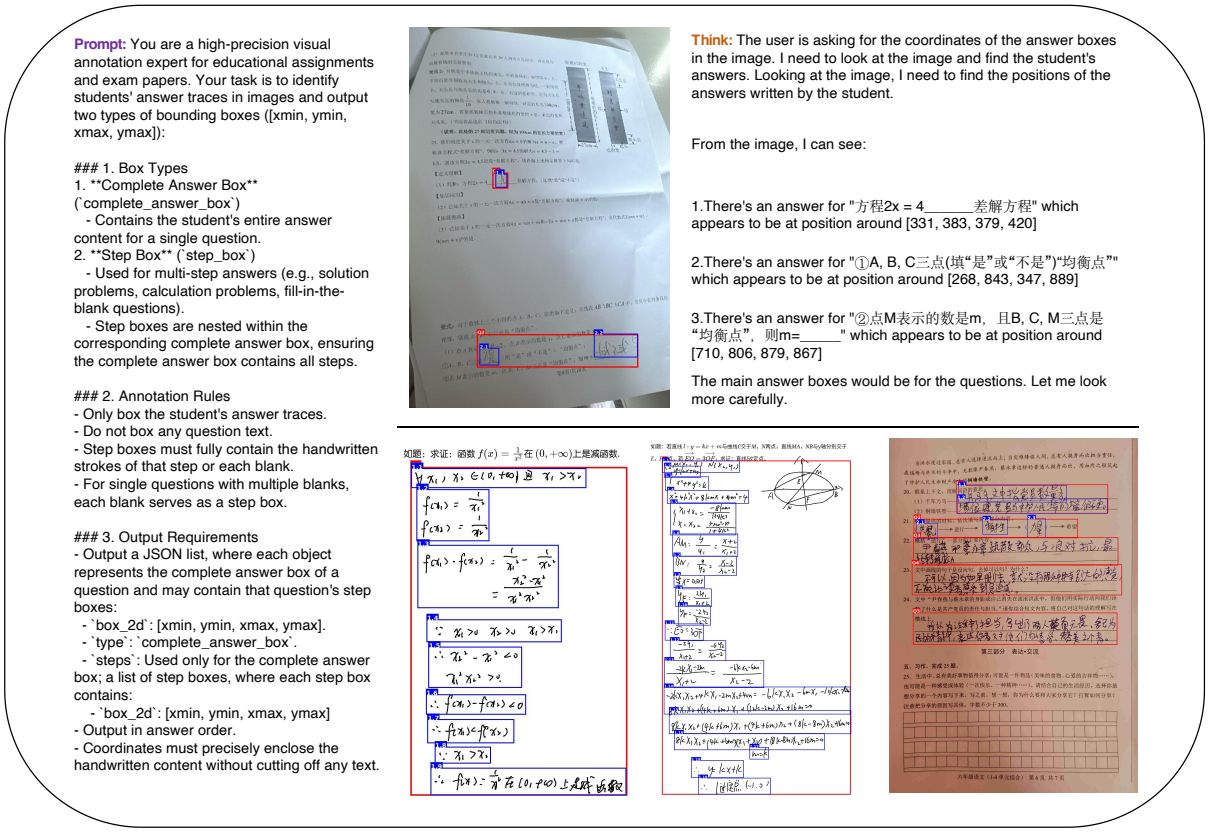


Figure 5: **Task overview and sample diversity on HG-Bench.** *Top*: the shared evaluation prompt, a representative two-page handwritten input with predicted question-level boxes (red) and step-level boxes (blue), and the model's textual grounding trace. *Bottom*: three additional samples spanning function derivation, calculation, and Chinese long-form writing, showing the benchmark's diversity in layout, handwriting density, and subject type. Additional failure-specific cases appear in Appendix E.

mode is consistent across families: page-boundary hallucinations (boxes attributed to the wrong page) and index shuffling (out-of-order step IDs across page breaks). The reference SFT system shows substantially smaller decay, indicating that exposure to multi-page context during SFT stabilizes cross-page coordinate mappings. We flag multi-page coordinate stability as an open problem that current cross-modal attention does not solve from scale alone.

7 Analysis

We categorize model failures into five types: **hallucinated boxes** covering no handwriting, **missed steps** where a multi-step answer receives only a question-level box, **page misalignment** with boxes assigned to the wrong page, **step over-/under-segmentation** where one step is split or multiple steps are merged, and **format failures** such as invalid JSON or schema violations. Closed-source frontier systems mainly fail through missed steps and under-segmentation, while weaker open-weight models additionally suffer from frequent format errors. Figure 5 illustrates the task setting and representative HG-Bench samples; detailed failure cases are provided in Appendix E.

7.1 Inter-Model Agreement

To measure how well HG-Bench separates model capabilities, we count, for each test sample, how many of the nine zero-shot baselines achieve question-level $F_1 \geq T$, using $T = 0.5$ as the standard matching threshold and $T = 0.7$ as a stricter high-quality bar.

HG-Bench is far from saturated. At $T = 0.5$, only $3/500$ samples (0.6%) are solved by all nine baselines, while 115 samples (23.0%) are missed by every baseline; the remaining 76% are solved by one to eight models, indicating a smooth difficulty gradient. At $T = 0.7$, the universally missed set rises to 44.6%, whereas the all-passed set remains 0.6%, confirming that current frontier VLMs leave substantial headroom.

Hard samples are not dominated by multi-page

inputs. Among the 115 universally missed samples at $T = 0.5$, 59 (51%) come from the enterprise channel and 56 (49%) from the consumer channel, closely matching the balanced test-set composition. Thus, title-level failures are not explained by page count alone; handwriting density and step decomposition remain the main residual challenges.

Targeted SFT recovers genuinely hard cases. On the 115 samples missed by every zero-shot baseline at $T = 0.5$, the reference SFT system reaches question-level $F_1 \geq 0.5$ on 68 samples (59.1%). Under the stricter $T = 0.7$ threshold, it still rescues 45.7% of the 223 universally missed samples. This shows that HG-SFT does not merely improve easy cases, but specifically recovers examples beyond the reach of frontier zero-shot systems, supporting the learnability claim in Sec. 6.3. This sample-level view is important because aggregate SFT performance can mask persistently unlearned subsets even after apparent convergence (Xue et al., 2026b).

8 Conclusion

We introduced HG-Bench, the first benchmark for per-question and per-step answer-region grounding on multi-page handwritten K–12 homework, with a page-aware protocol for imbalanced step-bearing pages. Evaluating frontier closed-source and open-weight VLMs shows that $\mathcal{F}_A$ plateaus around 50–55, while step-level grounding ($\mathcal{F}_S^{\mu}$) remains the main bottleneck, with several models falling below 10. A simple SFT reference based on an open-weight 9B model and ∼10k in-domain examples surpasses all evaluated closed-source systems without reinforcement learning, highlighting targeted supervision as a practical path toward structured homework grounding.

Limitations

HG-Bench targets Chinese K–12 homework; transfer to other languages and to higher-education work is not measured. The benchmark contains 500 samples, which limits the resolution of finer-grained subject- or grade-level conclusions. Evaluation uses a single $\mathrm{IoU} = 0.5$ matching threshold; performance under tighter or looser thresholds is not reported here. Step decomposition contains an irreducible subjective component, particularly for short solve-type answers, which the two-stage annotation protocol (Section 4) mitigates but does not eliminate. **Inter-annotator agreement** is reported on a 50-sample randomly sampled subset with two independent annotators (Section 4); extending IAA to a larger subset and reporting per-category and question/step breakdowns is a near-term release item. **Additional reference-system ablations** (channel and supervision-target ablations) are deferred to future work building on the released HG-SFT corpus; the reference SFT system is reported as a learnability lower bound rather than as a methods contribution. Finally, the reference system was trained without a reinforcement-learning stage. Model-internal uncertainty has recently been used to trigger more efficient generative reward modeling (Xue et al., 2026a), while latent-state scoring can select among multiple reasoning trajectories (Yu et al., 2025b); whether such post-training would further improve performance on HG-Bench remains open.

Ethics Statement

Source homework images were anonymized prior to inclusion in either the benchmark or the training pool. No personally identifying information about individual students appears in either resource, and no per-student attribute is exposed to the evaluated models. The intended downstream use of systems built on HG-Bench is teacher-side grading assistance, not automated student evaluation, ranking, or admissions decisions. We do not release individual student work; benchmark images are obtained through a commercial partnership with a third-party data provider under terms permitting educational research use, and we redistribute only derived annotation metadata together with anonymized identifiers. Any downstream deployment should pair this data minimization with infrastructure-level security controls. Policy-driven vulnerability assessment can aggregate heterogeneous risk factors to prioritize remediation (Shao, 2026a), while SHAP-based anomaly-detection ensembles provide an auditable approach to monitoring network traffic in security-critical embedded settings (Shao, 2026b).

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A Annotation Guidelines (Excerpts)

This appendix summarizes the core rules used by the annotator pool (Section 4). The full guideline document is released with the benchmark.

Skip rules. A page is *skipped* (no boxes drawn) when any of the following apply: (i) the image is corrupted, blank, or illegible; (ii) teacher grading marks are present anywhere on the page (check marks, crosses, written scores, marker corrections); or (iii) all student-produced marks on the page are non-conventional (free drawings, doodles, off-task content). Red-pen marks that are clearly part of the student's own answer (e.g., underlining or highlighting) are not treated as grading marks and the page is retained.

Two-level boxing. Every region of student handwriting is enclosed by an axis-aligned bounding box. Each question receives exactly one question-level `complete_answer_box` that tightly bounds all of the student's handwriting attributable to that question. For multi-step solutions (computation, derivation, multi-blank fill-in), an additional ordered list of `step_box` elements decomposes the answer; each step box must be fully contained within its parent question-level box. A single handwritten line may never be split across two boxes. Informal scratch work (calculations in the margin, crossed-out drafts) is not boxed.

Choice and judgment answers. When both a bubble-fill region and a hand-written letter answer are present, annotators box the bubble region. When only one is present, that one is boxed.

Tagging. Each question-level box carries (i) the parent title number and (ii) the sub-question number, separated by a slash. The original tags follow Chinese typographic convention (full-width parentheses, circled digits, blank-N identifiers); the released schema preserves the original strings verbatim. Step IDs are integers assigned in the order in which the student wrote the steps, starting from 1.

Hierarchical containment. The containment property—each step box contained in its parent question-level box—is enforced at annotation time and re-verified by the QC reviewer.

B Reference SFT System Details

This appendix provides the full configuration of the reference SFT system reported in Section 6.2. The system is intended as a learnability probe and reproducible lower-bound reference for HG-Bench, not as a state-of-the-art model submission.

Base checkpoint. We initialize from GLM-4.6V 9B (GLM-V Team, 2025), an open-weight vision–language model released by Z.ai on 2025-09-30. We deliberately avoid initializing from any newer same-family checkpoint, ensuring that gains over the baselines in Table 2 cannot be explained by a stronger or more recent foundation model.

Training data. We fine-tune on **HG-SFT**, the 9,920-sample training pool derived from the same annotation effort as the 500-sample HG-Bench test set. The two pools are disjoint by construction. As an additional safeguard, every training image is checked against the HG-Bench test pool using perceptual hashing (pHash, Hamming distance ≤ 5) together with exact metadata matching on user ID and capture timestamp. Before deduplication, the raw candidate training pool contained 14,264 samples; pHash-based filtering removed 4,344 near-duplicates, yielding the final $N_{\text{train}} = 9{,}920$. HG-SFT preserves the natural enterprise / consumer composition of the source pool (6,515/3,405), and we do not re-weight between channels.

Optimization recipe. We perform single-stage supervised fine-tuning for 3 epochs, corresponding to 930 optimization steps, with global batch size 32, sequence length 32,768, bfloat16 precision, and AdamW optimization. The learning rate follows a cosine schedule with 30 warmup steps, peak learning rate 1×10^{-6}, minimum learning rate 5×10^{-7}, weight decay 0.1, and gradient clipping at 1.0. Training runs on a single 8×H100-80GB node with tensor parallelism = 2, context parallelism = 2, and sequence parallelism enabled, including the ViT tower. Table 3 lists the complete hyperparameter and system configuration.

C Prompt Templates

Evaluation prompt. All baseline VLMs are queried with a single unified prompt template (English translation below; the verbatim Chinese version actually used in evaluation is released with the benchmark).

> You are a high-precision visual annotation expert for educational homework and exam papers. Your task is to identify each student's handwritten answer regions in the provided images and output two types of bounding boxes, with coordinates in

Item	Value
Base checkpoint	GLM-4.6V 9B (GLM-V Team, 2025)
Precision	bfloat16
Sequence length	32,768
Visual tokens / image	up to 10,000 (variable shape, jitter 0.75–1.25)
Optimizer	AdamW, $(\beta_1, \beta_2) = (0.9, 0.95)$, $\epsilon = 10^{-8}$
Weight decay	0.1
Gradient clipping	1.0
LR schedule	cosine, 30-step warmup
Peak / min LR	1×10^{-6} / 5×10^{-7}
Global batch size	32 (micro-batch 1)
Training steps	930 (= 3 epochs over 9,920 examples)
Dropout	0
Hardware	1 node, 8×H100-80GB
Tensor parallel	2
Context parallel	2
Sequence parallel	enabled, including ViT tower
Pipeline parallel	1
ZeRO / distributed optimizer	enabled, overlapped gradient reduction
Activation recomputation	full, per block
Activation offload	enabled, including ViT convolution and projection
Optimizer-state offload	CPU, 100%, precision-aware moments
Wall-clock time	≈ 3 hours
Total compute	≈ 24 GPU-hours

Table 3: Hyperparameters and system configuration of the reference SFT system.

`[xmin, ymin, xmax, ymax]` format normalized to `[0, 1000]`.

Box types. (a) `complete_answer_box`: tightly contains the student's entire answer to one question. For multiple-choice and true/false items, if both a bubble-fill region and a hand-written letter answer are present, prefer the bubble-fill region. (b) `step_box`: used for multi-step answers such as computation, derivation, and multi-blank fill-in items. Each step or blank is boxed separately and assigned a `step_id` starting from 1 in the order the student wrote them. Every step box must be nested inside the corresponding `complete_answer_box`.

Annotation rules. Box only the student's own marks (handwritten text, edits, ticks, connecting lines, drawings). Do not box printed question text or teacher corrections. Step boxes must fully contain the handwritten content of that step or blank. For a single multi-blank item, each blank becomes a separate step box in left-to-right, top-to-bottom order.

Output format. Emit a JSON list. Each element is one question-level object with the following fields: `box_2d` (the question-level box), `type` (fixed to `complete_answer_box`), and an optional `steps` list whose elements each carry their own `box_2d` and integer `step_id`. Items must be emitted in the order the student answered. Coordinates must tightly bound the handwriting without cropping any character.

A schematic example of the expected JSON output is shown in Figure 6.

Format-reminder retry prompt. When the format-tolerant parser fails to recover a valid JSON array from a model's first reply, a single retry is issued with the appended instruction:

> *Your previous response could not be parsed as a valid JSON array. Please reply with only the JSON array as described above, with no surrounding prose or markdown code fences.*

```
[
  {
    "box_2d": [100, 200, 180, 300],
    "type": "complete_answer_box"
  },
  {
    "box_2d": [400, 220, 490, 320],
    "type": "complete_answer_box",
    "steps": [
      {"box_2d": [410, 230, 440, 320], "step_id": 1},
      {"box_2d": [450, 230, 480, 320], "step_id": 2}
    ]
  },
  {
    "box_2d": [500, 220, 580, 780],
    "type": "complete_answer_box",
    "steps": [
      {"box_2d": [510, 230, 540, 780], "step_id": 1},
      {"box_2d": [550, 230, 580, 780], "step_id": 2},
      {"box_2d": [590, 230, 620, 780], "step_id": 3}
    ]
  }
]
```

Figure 6: Example JSON output illustrating the two-level box schema: one multiple-choice question with no steps, one fill-in item with two ordered blanks, and one solve item with three ordered derivation steps.

Persistent failures after this retry are recorded as terminal errors via the sentinel `FAIL_STR` (`success` = *False*).

D Supplementary Metrics

In addition to the two primary localization metrics ($\mathcal{F}_A$ and $\mathcal{F}_S^{\mu}$) reported in Section 5, we record four supplementary metrics in Table 2 to characterize the macro behavior of step decomposition and the parse-time reliability of each VLM.

Step-level macro F_1 ($\mathcal{F}_S^M$). A macro-aggregated complement to $\mathcal{F}_S^{\mu}$. Restricted to the subset of samples that contain at least one step box in ground truth, we compute the per-sample step F_1 (averaging per-page step F_1 within each sample) and then take the unweighted mean over all such samples. Compared with $\mathcal{F}_S^{\mu}$, which up-weights pages with denser step structure, $\mathcal{F}_S^M$ treats every step-bearing sample equally and therefore amplifies the contribution of short multi-step answers (single derivations, multi-blank fills). A model that does well only on long derivations but collapses on short multi-step items will show a larger gap between $\mathcal{F}_S^{\mu}$ and $\mathcal{F}_S^M$.

Parse success rate (Succ%). The fraction of the $N = 500$ samples on which the model's response (after at most one format-reminder retry) yields a structurally valid JSON array conforming to the prescribed schema. Samples whose response cannot be parsed are counted as failures and contribute

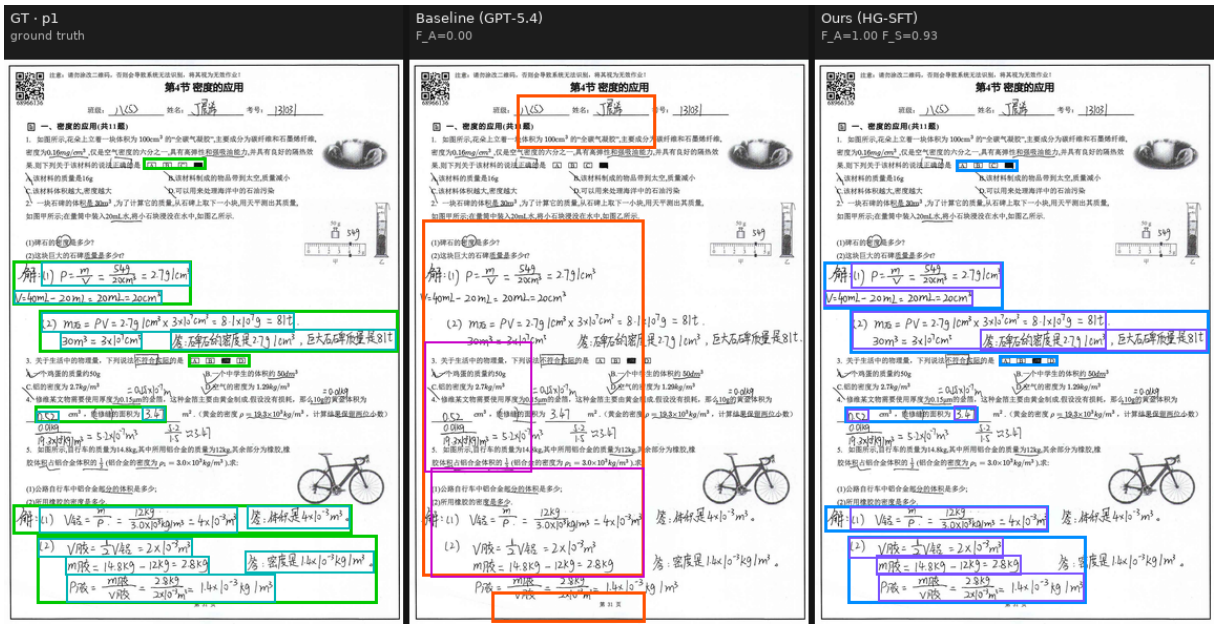


Figure 7: **Case 1: universally-missed sample (universal rescue).** One of the 115 samples on which no zero-shot baseline reaches title-level $F_1 \geq 0.5$ (Section 7.1). Every closed-source frontier API and every open-weight baseline produce highly fragmented or mis-localized boxes; the reference SFT system recovers both the question-level answer regions and the ordered step decompositions.

0 to all localization metrics in the unified score $\bar{\mathcal{S}}$ below.

Unified score over all samples ($\bar{\mathcal{S}}$). A reliability-aware composite that combines $\mathcal{F}_A$ and $\mathcal{F}_S^{\mu}$ over the *entire* 500-sample test set: failed-parse samples contribute 0 rather than being excluded. Concretely, $\bar{\mathcal{S}}$ is the average of the per-sample composite (a weighted combination of question-level and step-level page F1, identical to the per-page reward used during evaluation) over all 500 samples. Comparing $\bar{\mathcal{S}}$ against $\mathcal{F}_A$ exposes the practical cost of format failures: a model with high $\mathcal{F}_A$ but low Succ% will see a sharp drop in $\bar{\mathcal{S}}$ (most clearly visible in the Doubao-Seed-2.0-Pro `260215` release, where $\mathcal{F}_A = 52.65$ but $\bar{\mathcal{S}} = 21.22$ because nearly half of outputs failed to parse).

Repetition rate (Rep%). The fraction of the 500 outputs in which the model produced a repeating textual or structural pattern (consecutive duplicate boxes, looping JSON fragments, or repeated content blocks detected by a longest-common-substring heuristic on the raw model response). Lower is better. Rep% is reported in Table 2 as a transparency signal; samples flagged as repetitive are still scored under the standard protocol and are not excluded from $\mathcal{F}_A$, $\mathcal{F}_S^{\mu}$, $\mathcal{F}_S^{M}$, Succ%, or $\bar{\mathcal{S}}$.

E Additional Qualitative Cases

This appendix complements the qualitative comparison in Section 7 (Figure 5) with four additional HG-Bench samples that exercise different failure modes and difficulty axes.



Figure 8: **Case 2: multi-page enterprise sample.** An enterprise answer sheet spanning two pages, illustrating page-misalignment and index-shuffling failures: several baselines attribute boxes to the wrong page index or emit step IDs out of writing order across the page break (cf. Section 6.4). The reference SFT system preserves page-correct attribution and the per-step ordering across pages.

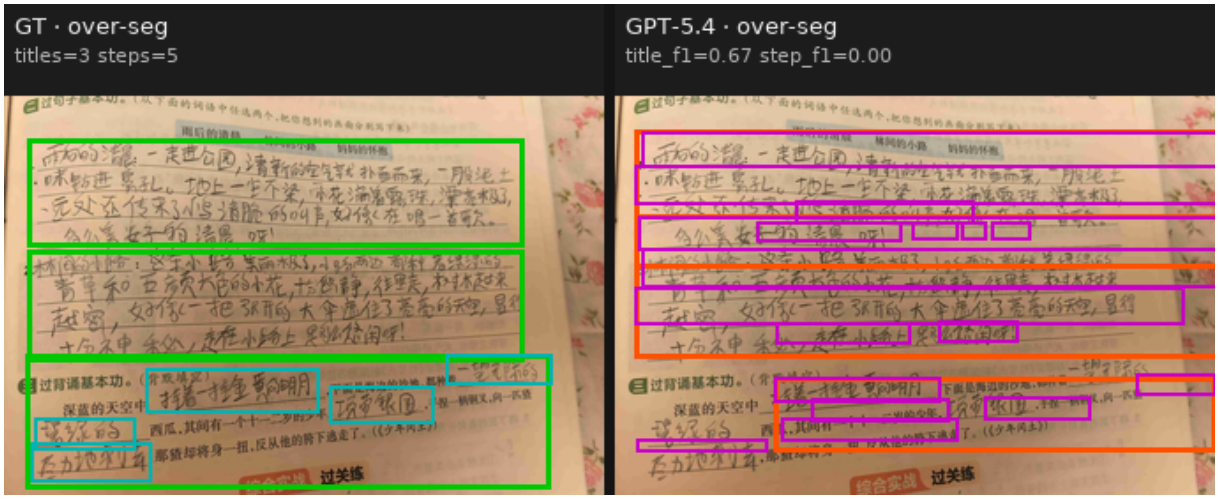


Figure 9: **Case 3: over-segmentation of step boxes.** A single multi-step derivation is split into too many step boxes by several baselines (e.g., each "=" or arithmetic operator gets its own box), inflating the step count and confusing downstream per-step grading. The reference SFT system produces step boxes whose count and granularity match the human annotation.

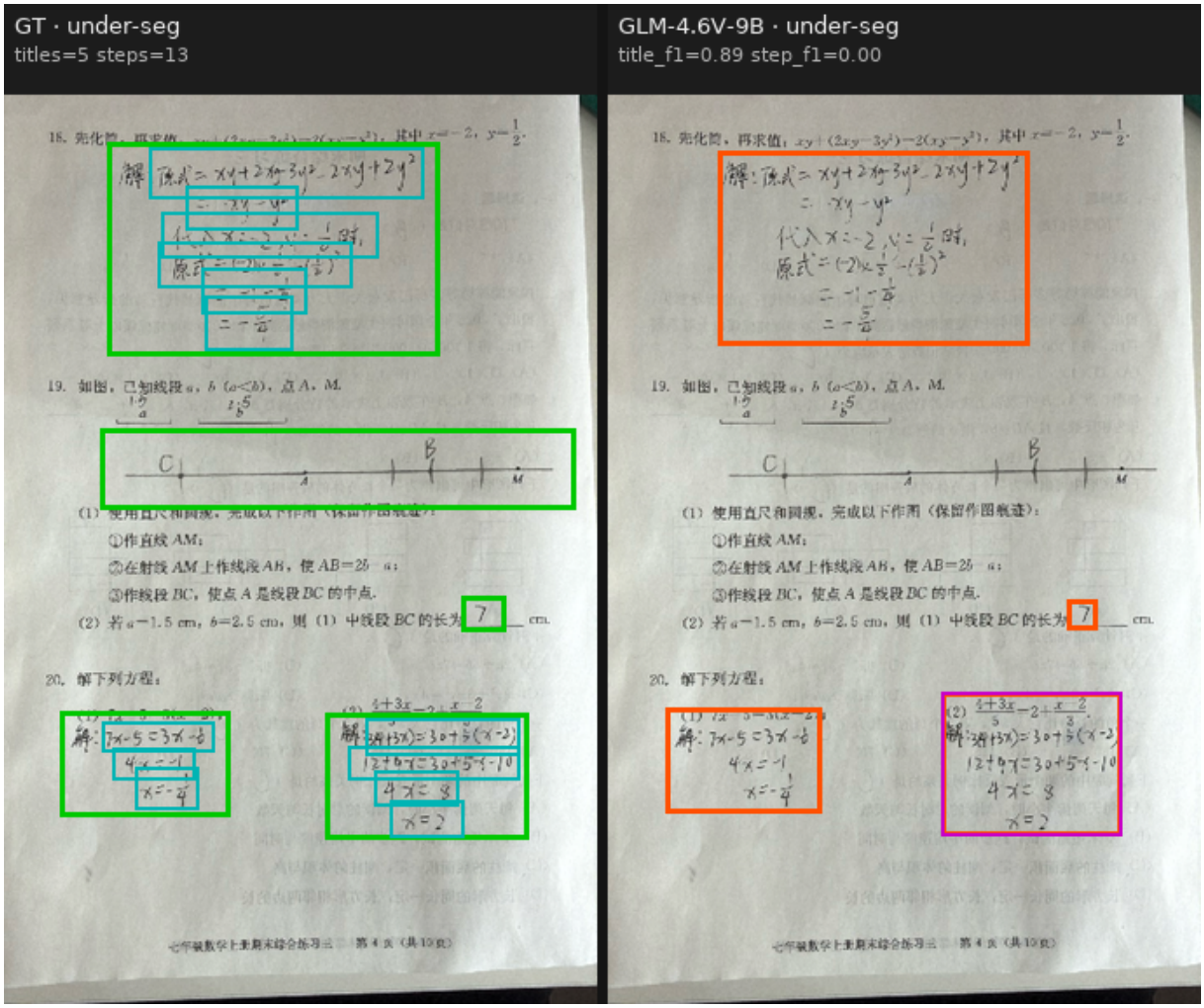


Figure 10: **Case 4: under-segmentation of step boxes.** The complementary failure to Case 3: several distinct derivation steps are merged into one large box, losing the per-step grading signal even when the overall question-level box is roughly correct. Closed-source baselines exhibit this pattern most often on dense multi-line solutions. The reference SFT system preserves the per-step boundaries.